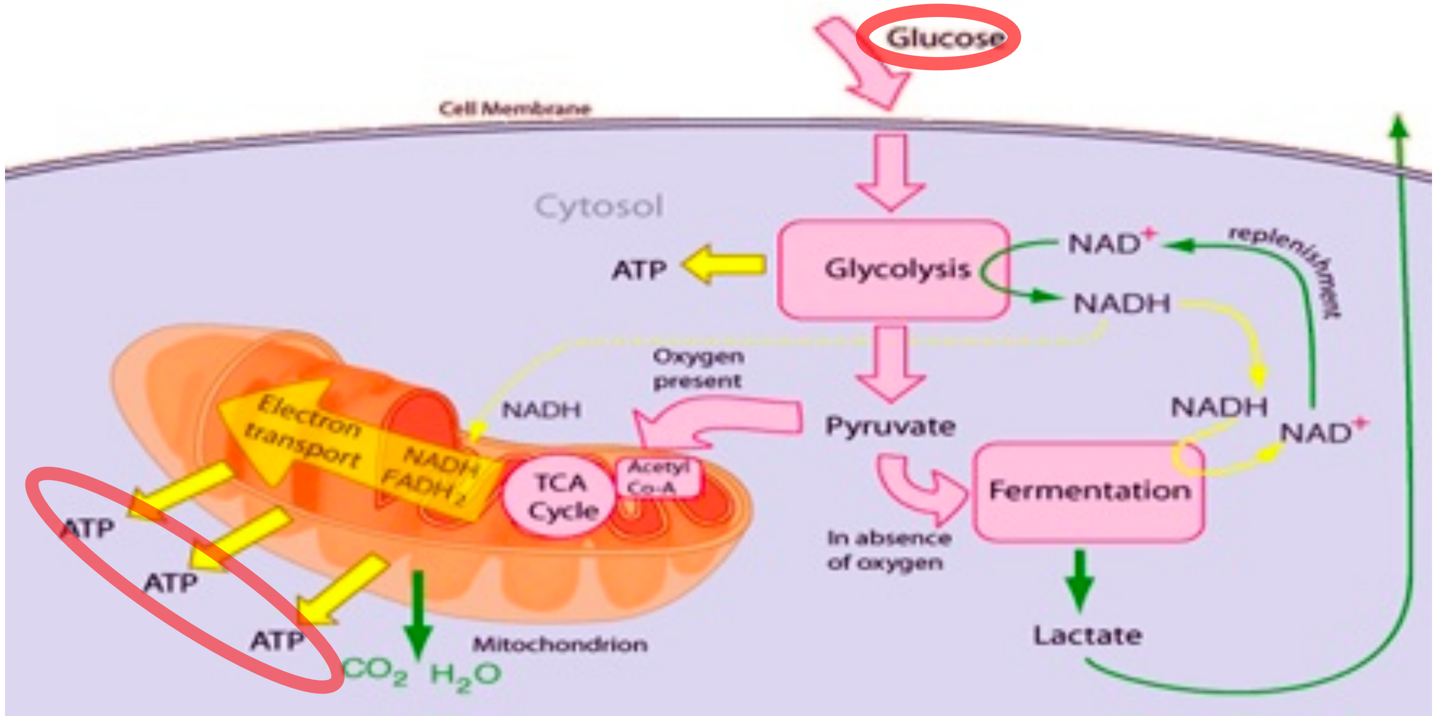
Glucose Metabolism (cont…)



https://www.assignmentpoint.com/science/health/describe-about-glucose-metabolism.html

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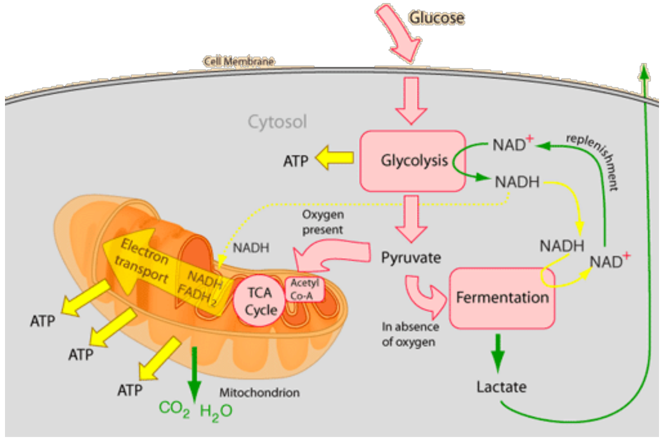


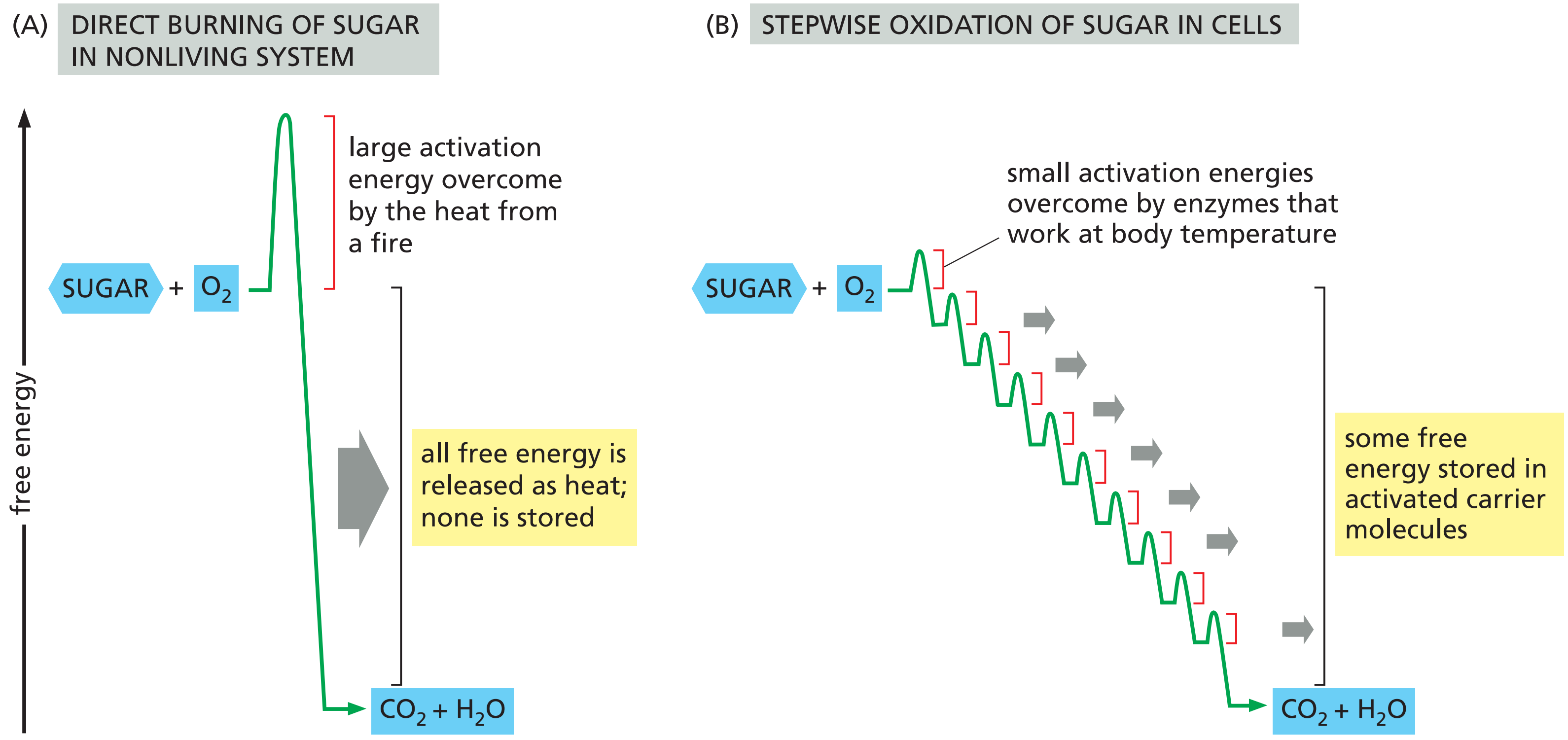
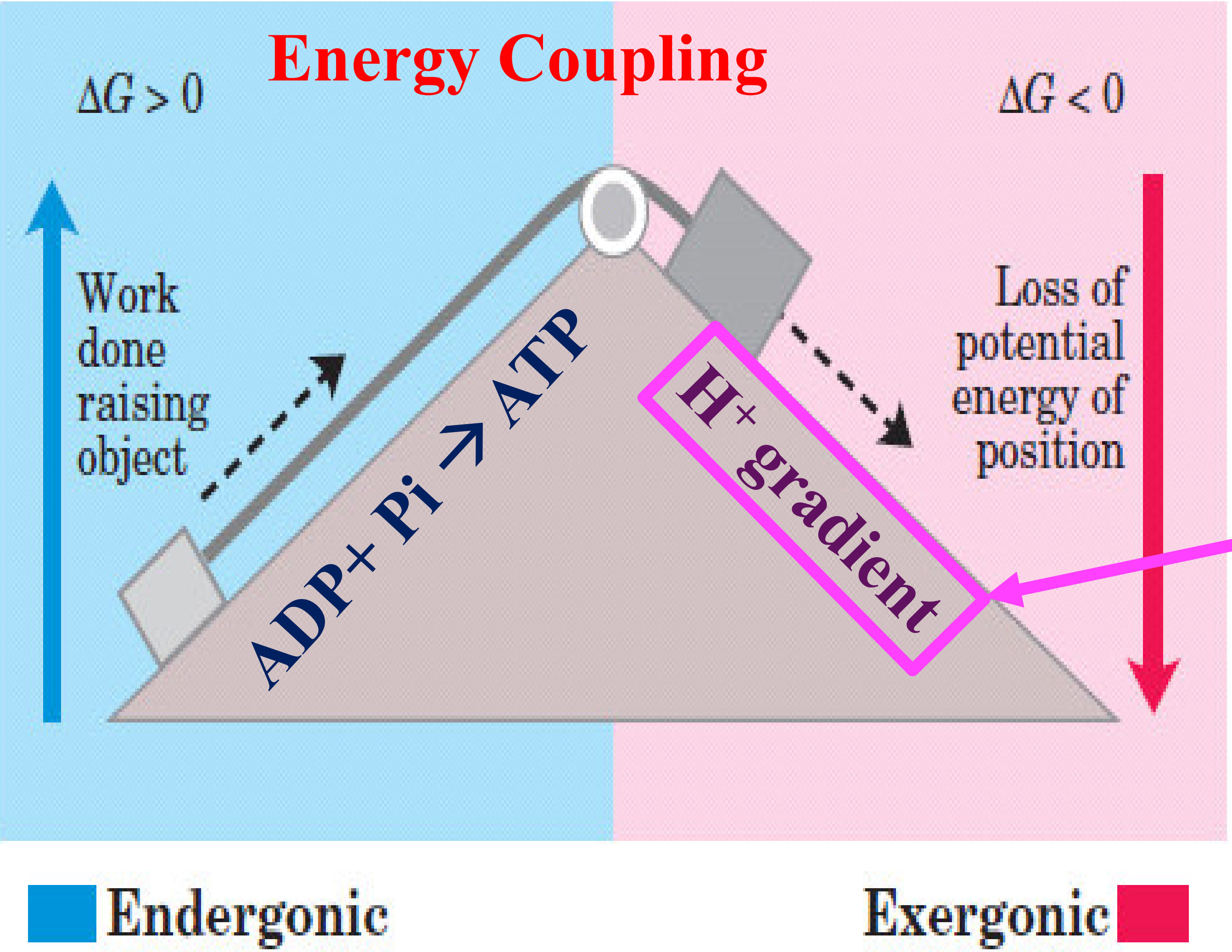
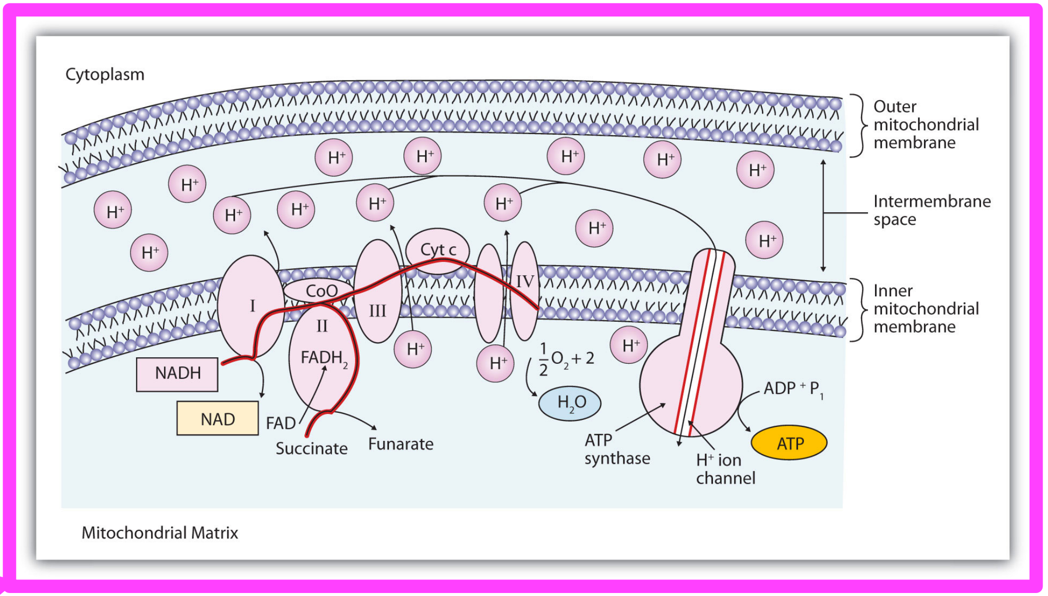


Figure 2–45 Schematic representation of the controlled stepwise oxidation of sugar in a cell, compared with ordinary burning. (A) If the sugar were oxidized to CO_2 and H_2O in a single step, it would release an amount of energy much larger than could be captured for useful purpose (B) In the cell, enzymes catalyze oxidation via a series of small steps in which free energy is transferred in conveniently sized packets to carrier molecules—most often ATP and NADH. At each step, an enzyme controls the reaction by reducing the activation-energy barrier that has to be surmounted before the specific reaction can occur. The total free energy released is exactly the same in (A) and (B).





ATP is uphill. Thus, ATP production from ADP+Pi require energy (from food).

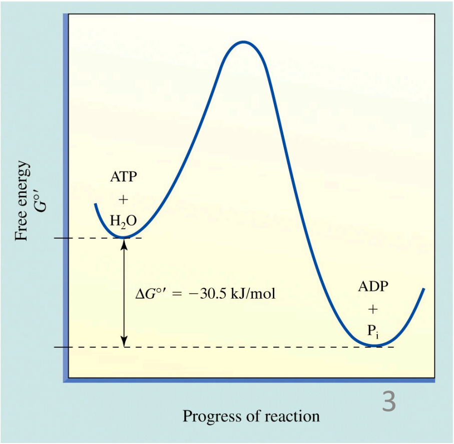
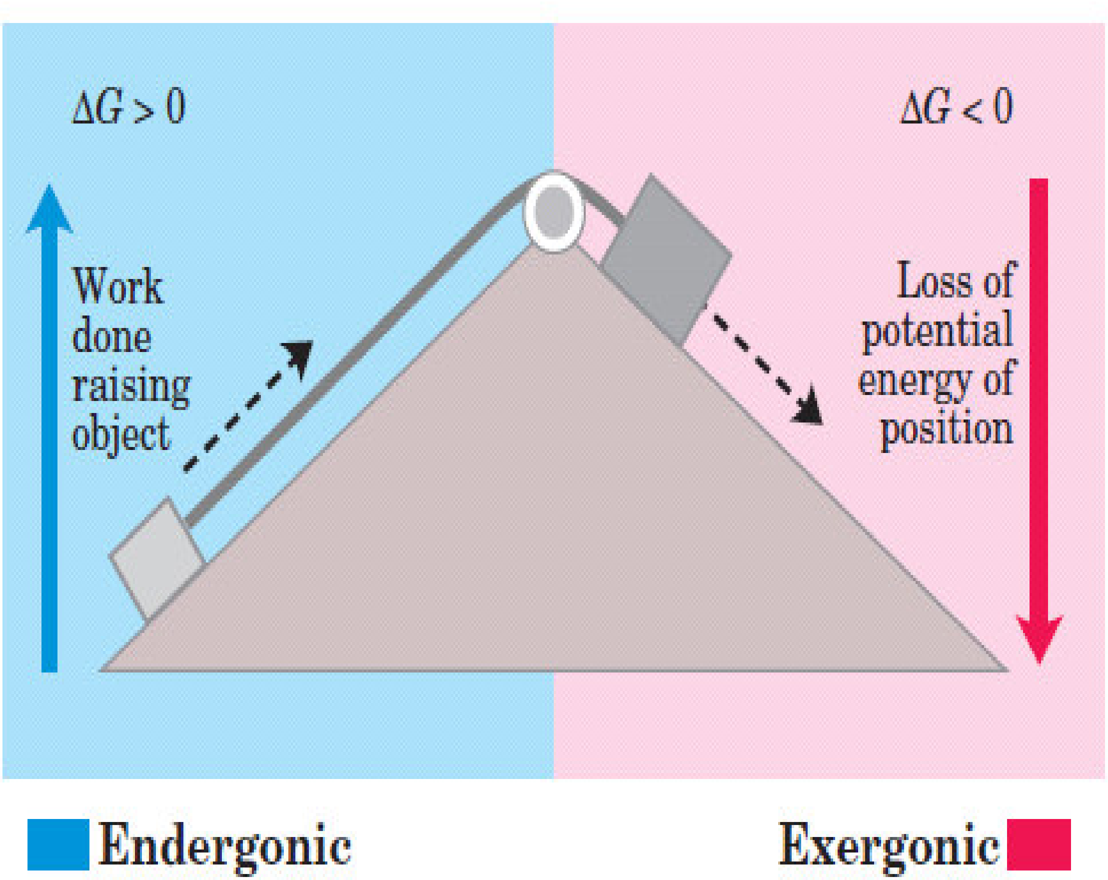


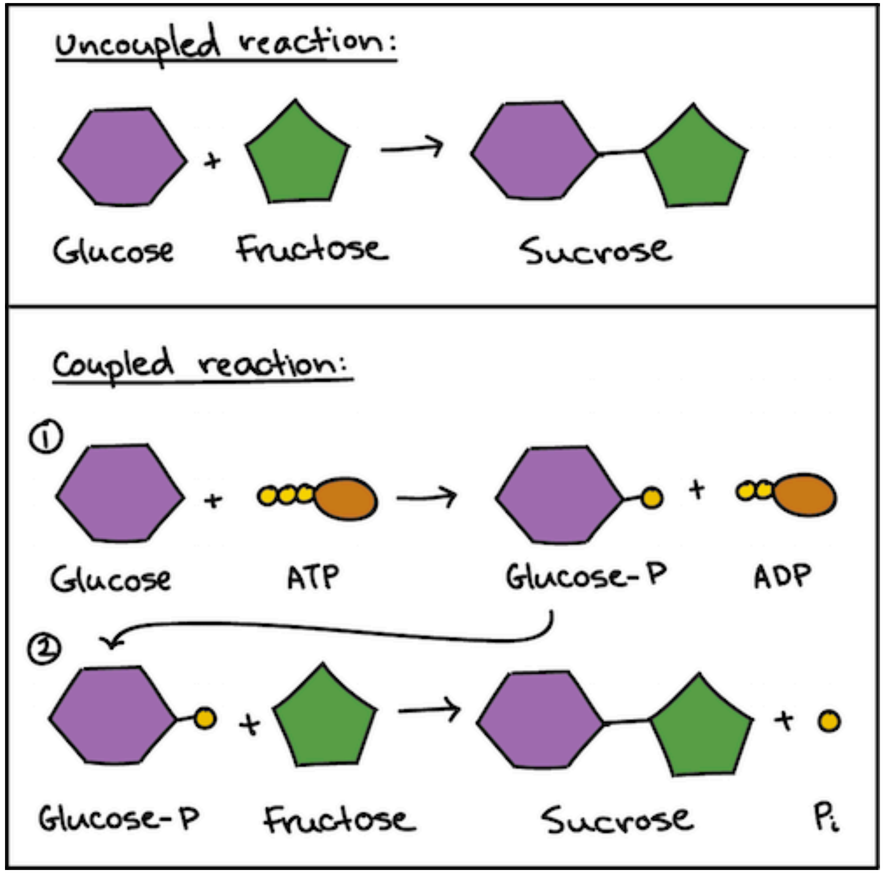
Figure 14-13 Concepts in Biochemistry, 3/e

Energy Coupling:

Life is all about energy coupling

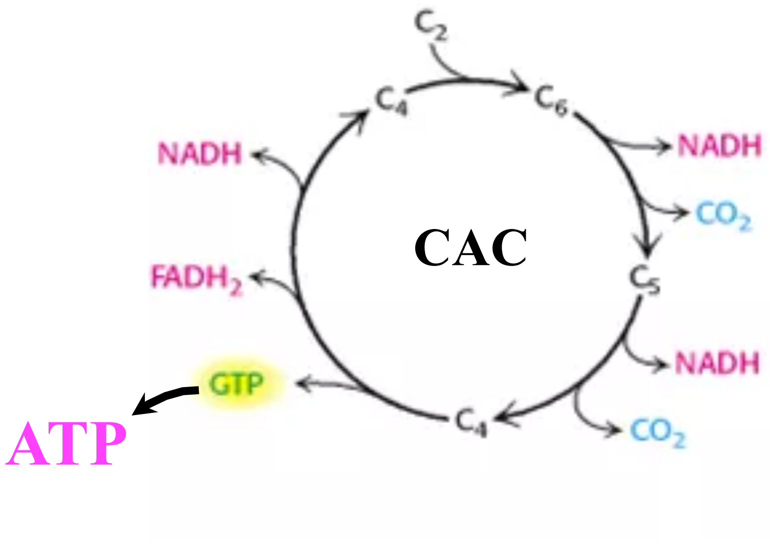


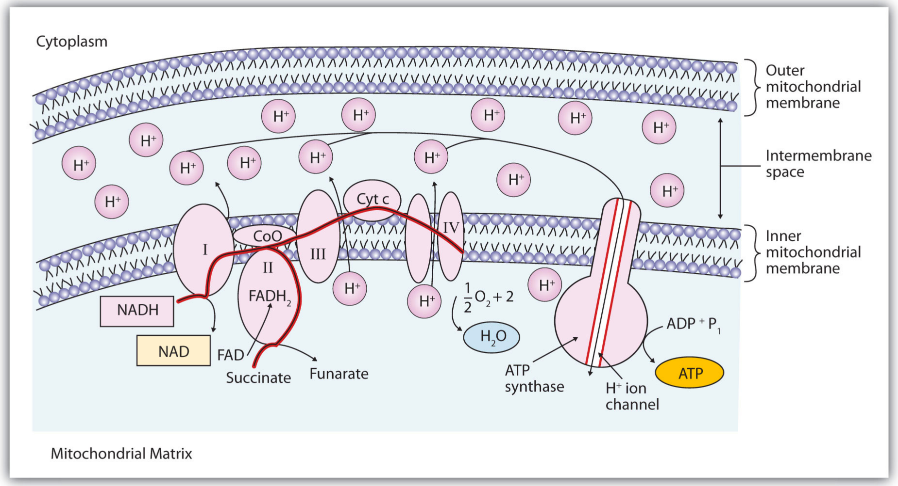
How can I couple reactions ?



Ans: Use the "product" of one reaction as a "reactant" of the other.

How much energy (ATP) is produced per Acetyl CoA ?





ETC

- ATP = 1
- NADH = 3
- $FADH_2$ = 1

- 1 NADH ➔ 3 ATP
- 1 $FADH_2$ ➔ 2 ATP

Total ATP production = 1+ 3x3 + 1x2 = 12 ATP (per Acetyl coA)

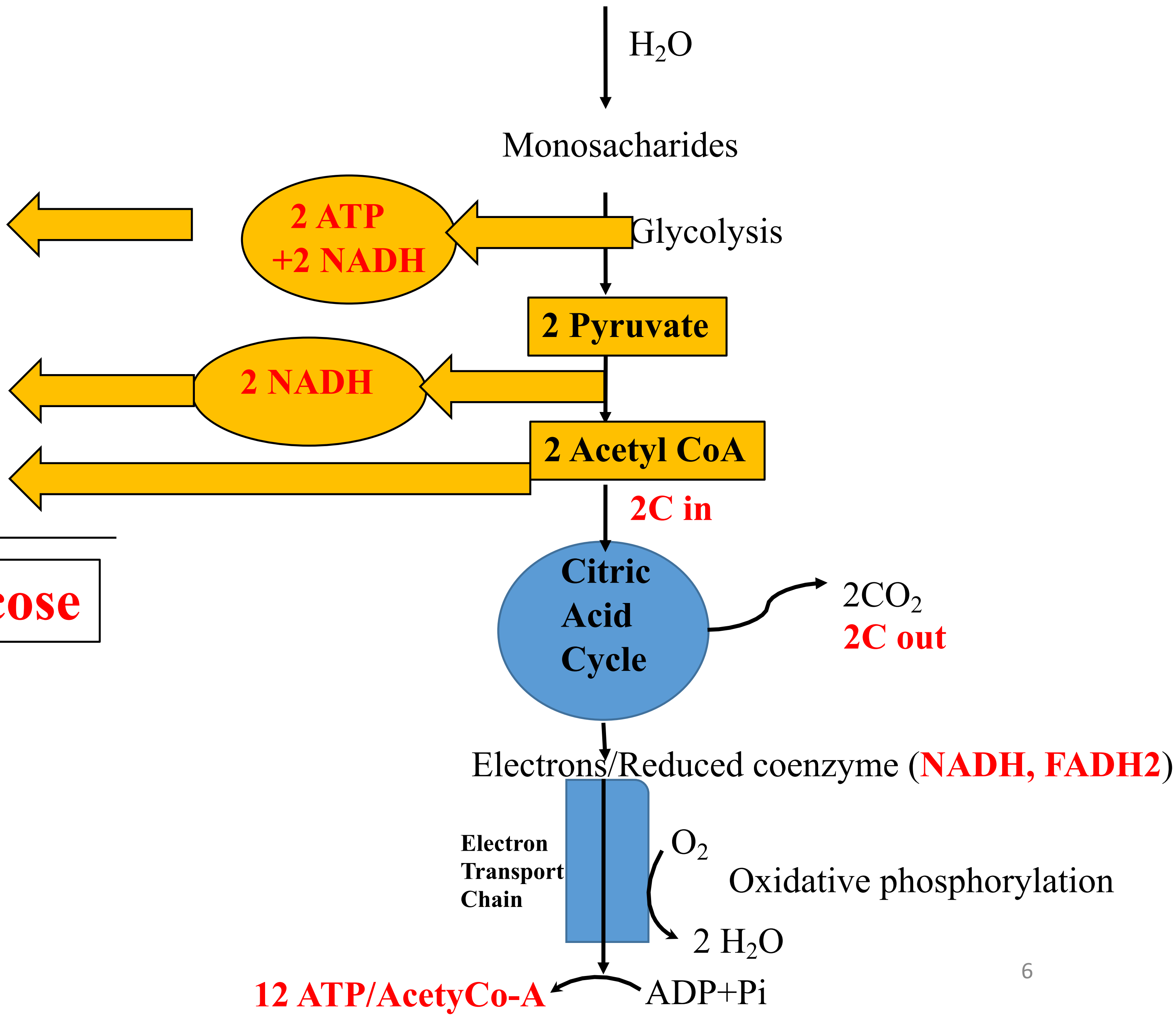
How much energy (ATP) is produced per glucose molecule ?

2 ATP
+2 x 3 ATP

+2 x 3 ATP

+2 x 12 ATP

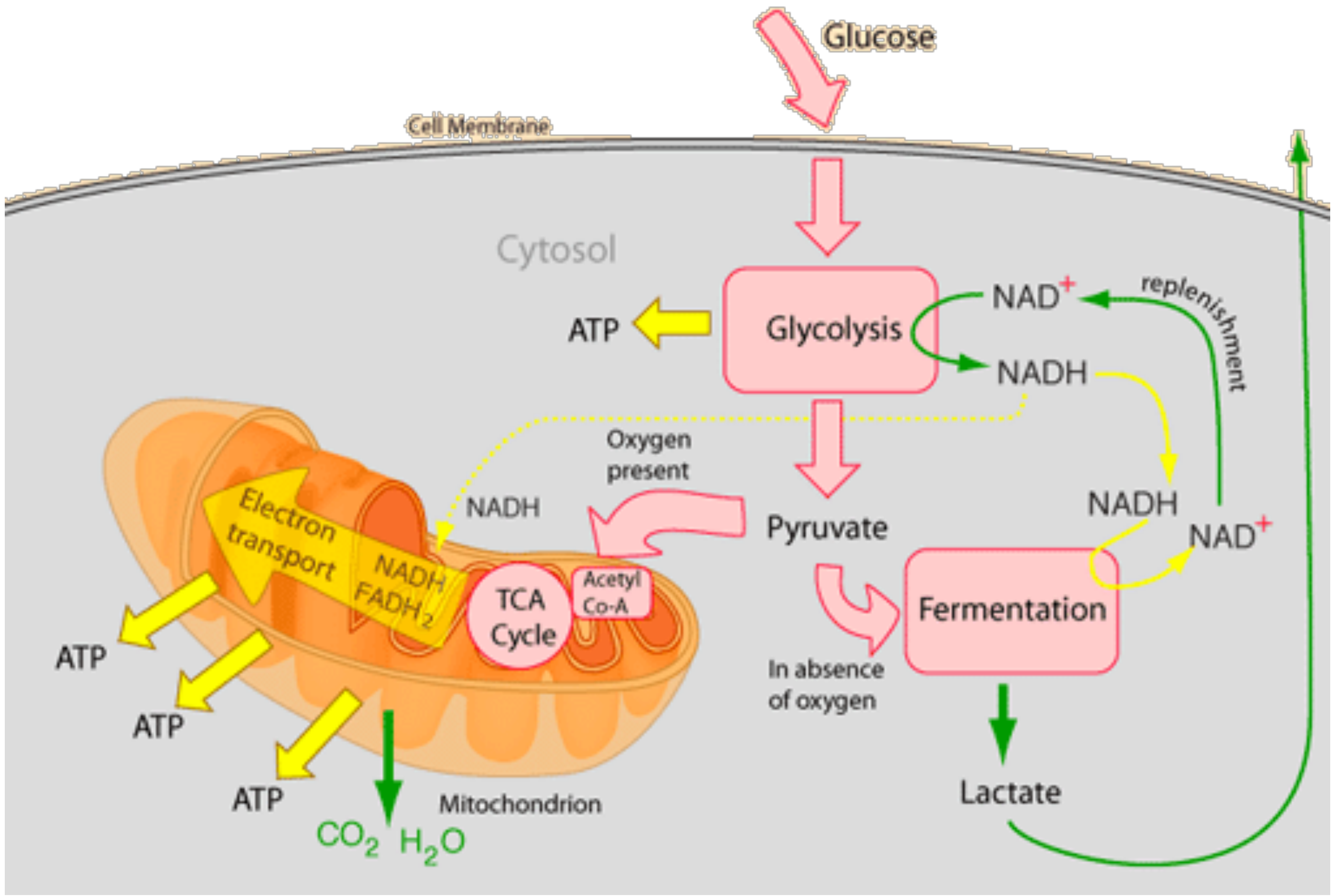
$\boxed{\textbf{38 ATP/ glucose}}$



Glucose = $C_6H_{12}O_6$

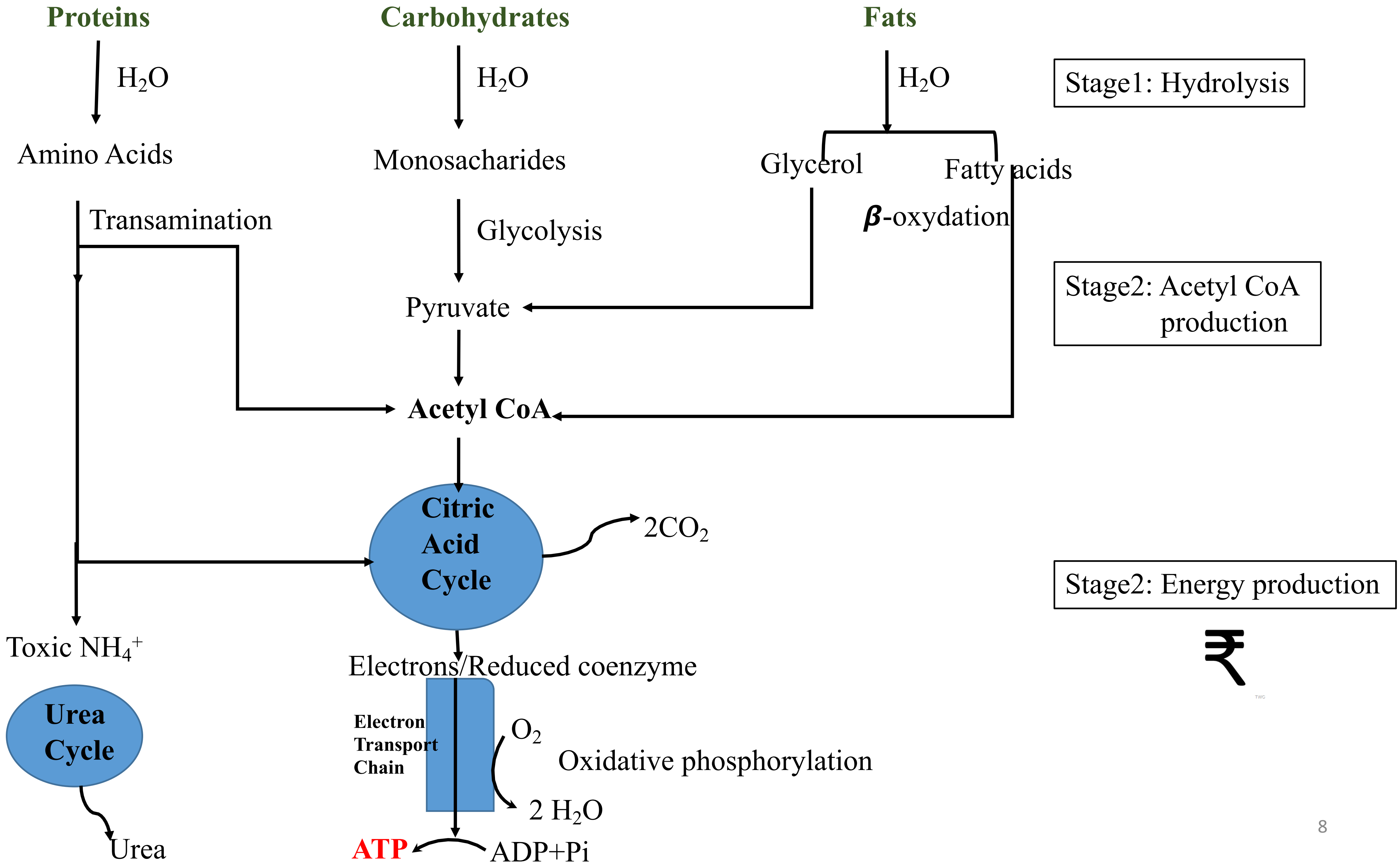
38/6 = 6.3 ATP/ C in glucose

How much energy (ATP) is produced per glucose molecule (For anaerobic condition) ?

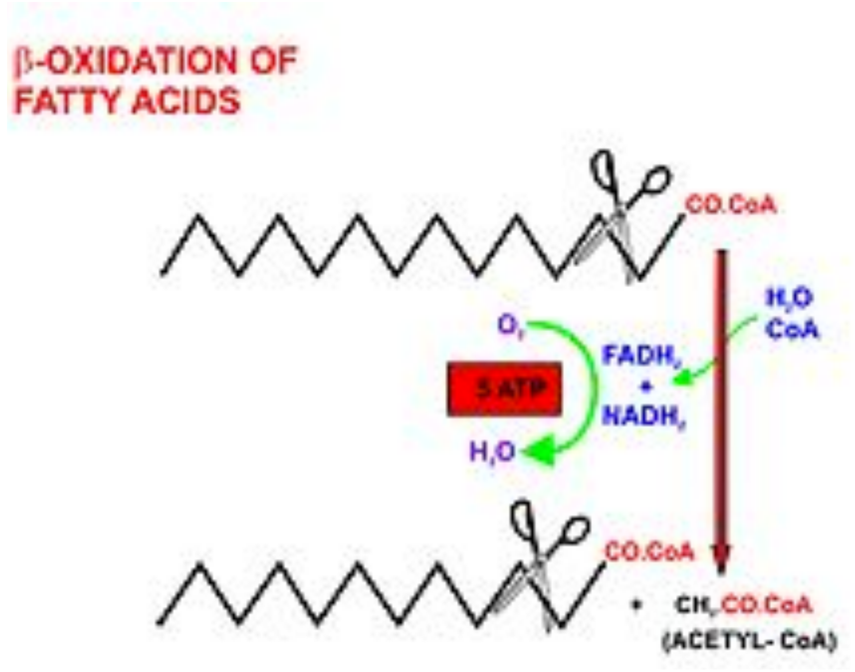


ANSWER: 2 ATP

(Note NADH is used up for Fermentation)


Proteins
Carbohydrates
Fats
H_2O
H_2O
H_2O
Stage1: Hydrolysis
Amino Acids
Monosacharides
Glycerol
Fatty acids
Transamination
Glycolysis
β-oxydation
Stage2: Acetyl CoA production
Pyruvate
Acetyl CoA
Citric Acid Cycle
$2CO_2$
Stage2: Energy production
Toxic NH_4^+
Electrons/Reduced coenzyme
Urea Cycle
Electron Transport Chain
O_2
Oxidative phosphorylation
2 H_2O
Urea
ATP
ADP+Pi

How much energy (ATP) is produced from FAT ?



Fats & oils

↓ H_2O Stage1: Hydrodysis

Fatty acids (EVEN NUMBER OF carbon atoms)

↓ ***β***-oxydation

#Each turn
1 FADH2
1 NADH
1 Acetyl coA

 # Last Turn
2 Acetyl coA

Cycles of β-Oxidation

The length of a fatty acid:

- Determines the number of oxidations and
- The total number of acetyl CoA groups.

Carbons in Fatty Acid	Acetyl CoA (C/2)	β-Oxidation Cycles (C/2 –1)
12	6	5
14	7	6
16	8	7
18	9	8

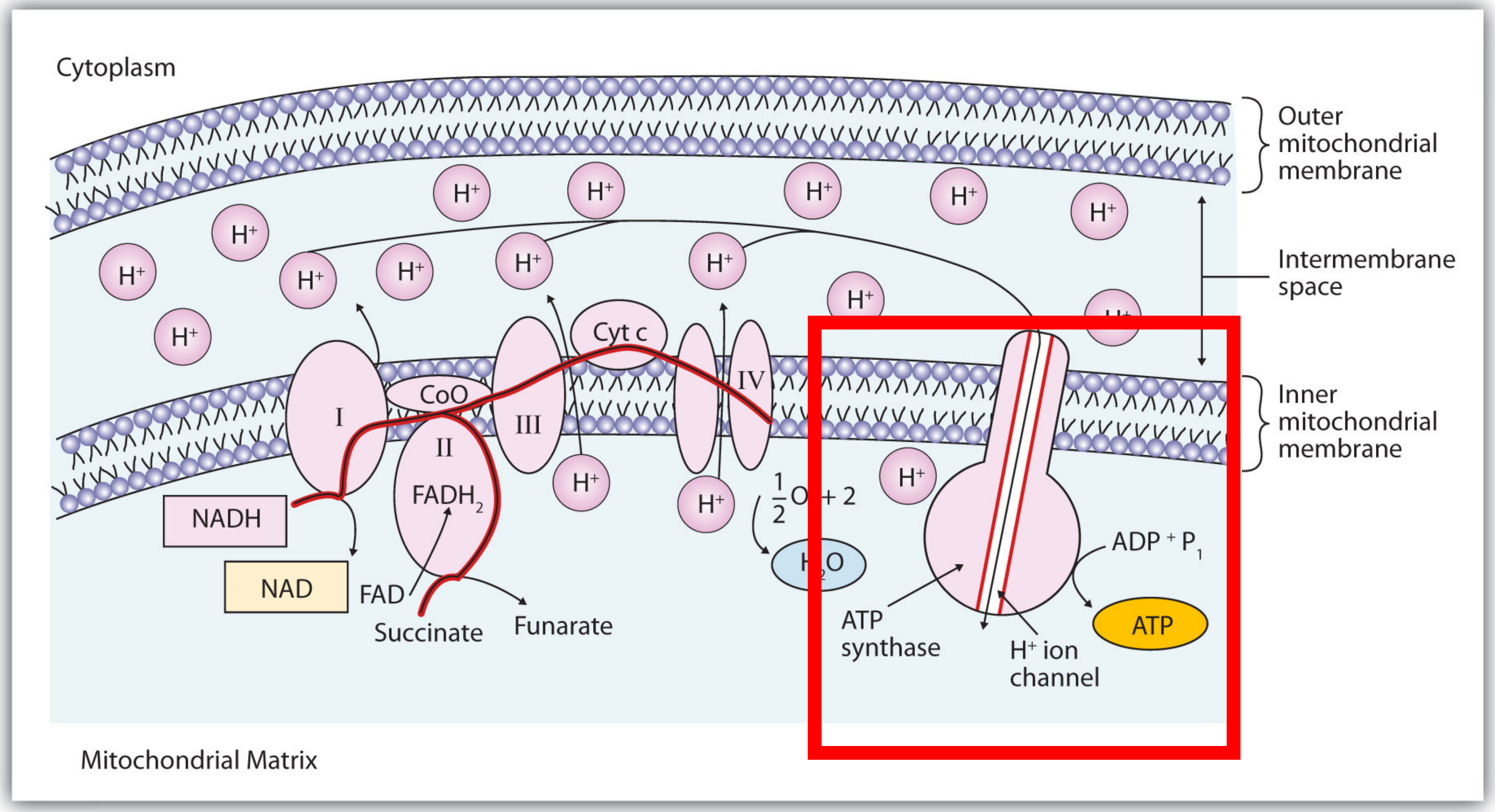
ATP for Lauric Acid C_{12}

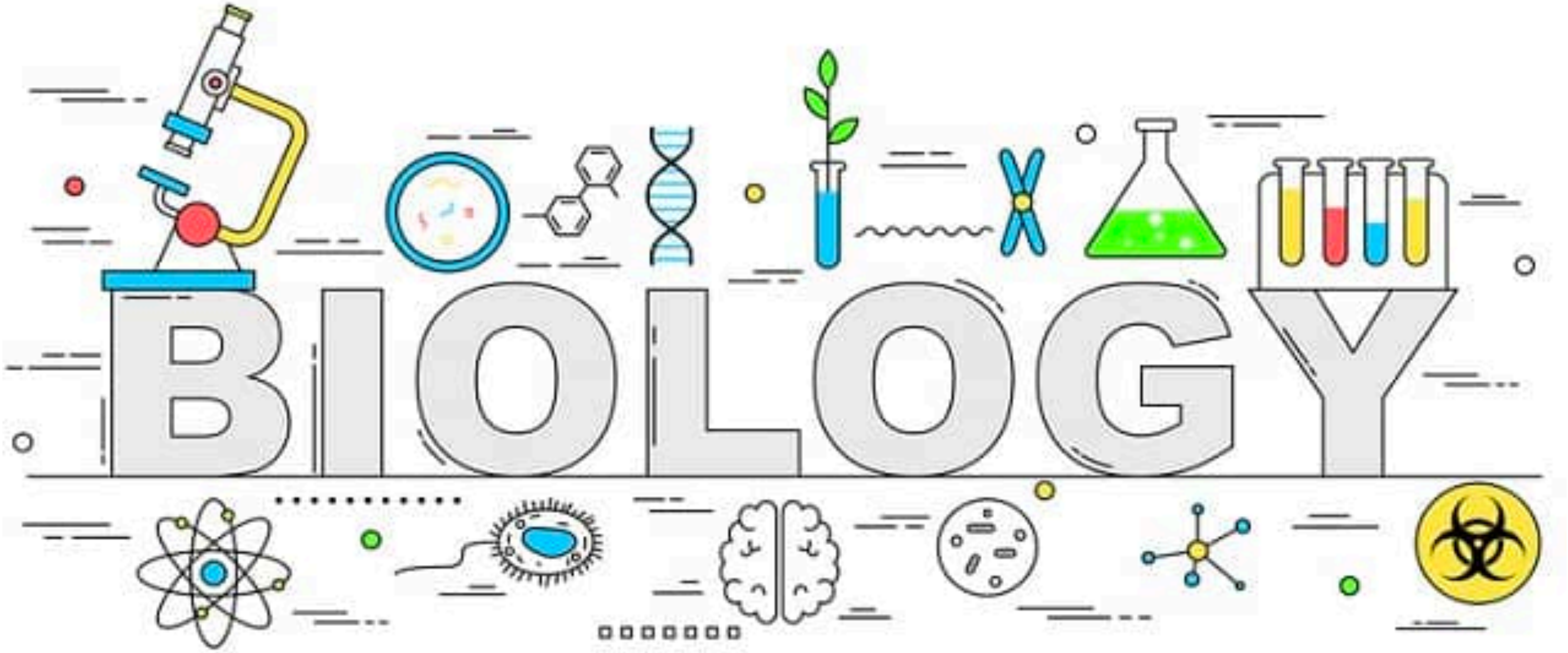
ATP production for lauric acid (12 carbons):

Activation of lauric acid	-2 ATP
6 Acetyl CoA	
6 acetyl CoA x 12 ATP/acetyl CoA	72 ATP
5 Oxidation cycles	
5 NADH x 3ATP/NADH	15 ATP
5 $FADH_2$ x 2ATP/$FADH_2$	10 ATP
Total	95 ATP

95/12 = 7.92 ATP/ C in fat

Fats gives more energy than glucose


Cytoplasm
Outer mitochondrial membrane
Intermembrane space
Inner mitochondrial membrane
H+
Cyt c
CoO
I
II
III
IV
NADH
NAD
FADH2
FAD
Succinate
Funarate
1/2 O + 2
H2O
ATP synthase
H+ ion channel
ADP + P1
ATP
Mitochondrial Matrix



Next: ATP synthase